MODULE-2

Regular Expression

→ method to represent a language

→ Let 'R' be a regular expression over alphabet Σ if R is :

1. ϵ, is regular expression denoting the set $\{\epsilon\}$
2. ϕ, is regular expression denoting the empty set $\{\}$
3. For each symbol $a \in \Sigma$, a is regular expression denoting set $\{a\}$
4. union of two RE is also Regular
5. Concatenation of two RE is also regular
6. Kleene closure * of RE is also regular
7. if R is regular, (R) is also regular
8. Nothing else, repeat 1 to 7 recursively

$R = \epsilon \quad L(R) = \{\epsilon\}$

$R = \phi \quad L(R) = \{\}$

$R = a \quad L(R) = \{a\}$

$R_1 \cup R_2$ = regular

$R_1 R_2$ = regular

a^* = regular $\{\epsilon, a, aa, aaa \ldots\}$

$(RE)^*$ = regular

Language accepted by FA is called Regular Language.
Regular Language is represented by Regular Expression

Regular expression for infinite language

Q Consider alphabet $\Sigma = \{a, b\}$

a) all strings having a single 'b'

$a^* b a^*$

here $a^* = \{\epsilon, a, aa, aaa --\}$

b) all string having atleast one 'b'

$(a+b)^* b (a+b)^*$

$(a+b)^* = \{a, \epsilon, b, aa, bb, aaa, ab ---\}$

c) all strings having bbbb as substring

$(a+b)^* bbbb (a+b)^*$

d) all strings end with 'ab'

$(a+b)^* ab$

e) all string start with 'ba'

$ba(a+b)^*$

f) all string begining & ending with 'a'

$a(a+b)^* a$

g) all string containing 'a'

$(a+b)^* a (a+b)^*$

h) all strings starting & ending with different symbol

$a(a+b)^* b$

or

$b(a+b)^* a$

RE is $a(a+b)^* b + b(a+b)^* a$

(i) all string having 2'b'

$a^* b a^* b a^*$

Q Find regular expression ~~or~~ langugi for $\Sigma = (a, b)$ having

a) no string

$RL = \{\}$

$RE = \phi$

b) length 0

$RL = \{\epsilon\}$

$RE = \epsilon$

c) length 1

$RL = \{a, b\}$

$RE = (a+b)$

d) length 2

$RL = \{aa, ab, ba, bb\}$

$$RE = (aa+ab+ba+bb)$$
$$= a(a+b) + b(a+b)$$
$$= (a+b)(a+b)$$

e) length 3

$RL = \{aaa, bbb, aba, aab, abb, baa, \ldots$

$RE = (a+b)(a+b)(a+b)$

f) atmost 1

$RL = \{\epsilon, a, b\}$

$RE = (\epsilon+a+b)$

g) atmost 2

~~RE = {~~

$RE = (\epsilon+a+b)(\epsilon+a+b)$

h) not more than 2 b's & 1 a

$L = \{\epsilon, a, b, ab, ba, abb, bab, bba$

$E = (\epsilon+a+b+ab+ba+ab+\ldots$

Identities for RE

Two RE, P & Q are equivalent ($P = Q$) iff P represents the same set of strings as Q does.

Let P, Q & R are RE then identity rules are :-

1. $\epsilon R = R\epsilon = R$
2. $\epsilon^* = \epsilon$
3. $(\phi)^* = \epsilon$
4. $\phi R = R\phi = \phi$
5. $\phi + = R = R$
6. $R + R = R$
7. $RR^* = R^*R = R^+$
8. $(R^*)^* = R^*$
9. $\epsilon + RR^* = R^*$
10. $(P + Q)R = PR + QR$
11. $(P+Q)^* = (P^*Q^*) = (P^* + Q^*)^*$
12. $R^*(\epsilon + R) = (\epsilon + R)R^* = R^*$
13. $(R + \epsilon)^* = R^*$
14. $\epsilon + R^* = R^*$
15. $(PQ)^*P = P(QP)^*$
16. $R^*R + R = R^*R$

Equivalence of two RE

Arden's Theorem helps in checking the equivalence of 2 RE.

Arden's Theorem

Let P & Q be two RE over input set Σ
The ~~RE~~ regular expression R is given as

$$R = Q + RP$$

has a unique solution

$$R = QP^*$$

Proof Let P & Q are two RE over Σ
If P does not contain ϵ then there exists R such that

$$R = Q + RP \rightarrow (1)$$

Replace R by QP^* in (1)

$$R = Q + QP^*P$$
$$R = Q(\epsilon + P^*P)$$
$$R = QP^* \quad \text{since } \epsilon + R^*R = R^*$$

Now to prove $R = QP^*$ is unique solution we will replace LHS of eq. (1) by $Q+RP$

~~R = Q+R~~

$$Q + RP = Q + (Q+RP)P$$
$$= Q + QP + RP^2$$

Again replace R by $Q+RP$

$$= Q + QP + (Q+RP)P^2$$
$$= Q + QP + QP^2 + RP^3$$

If we keep on replacing R by $Q+RP$, then

$$Q + RP = Q + QP + QP^2 + QP^3 + \ldots QP^i + RP^{i+1}$$
$$= Q(\epsilon + P + P^2 + \ldots P^i) + RP^{i+1}$$

Now from (1),

$$R = Q + RP$$

$$R = Q(\epsilon + P + P^2 + \dots P^i) + RP^{i+1} \rightarrow (2)$$

where $i \geq 0$

$$R = Q(P^*) + RP^{i+1}$$

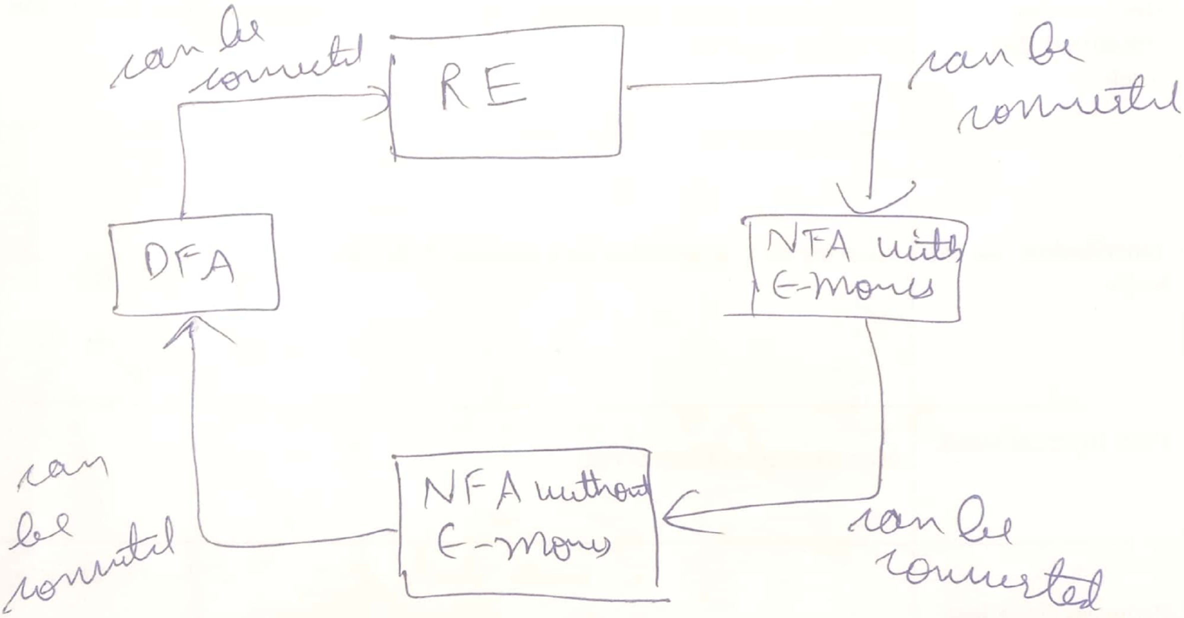
Let w be a string of length i

$$R = QP^*$$

LHS = RHS

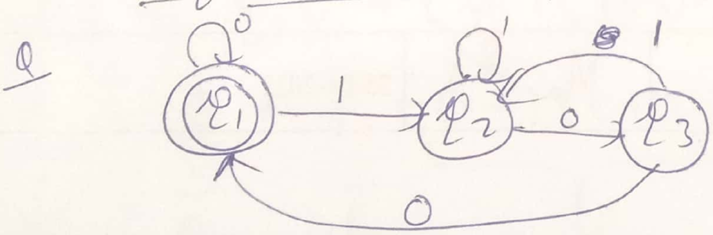
Hence proved

Relationship b/w FA & RE



Equivalence of DFA & RE

Q



Step 1 → For each state Q1, Q2, Q3, ... all edges that comes into state are written in equation format.

Step 2 → For initial state, add ϵ

Step 3 → calculate all eq.

Step 4 → Result is value of final state

Ans

$$q_1 = \underset{\text{incoming edge}}{q_1 0} + \underset{\text{incoming edge}}{q_3 0} + \epsilon \rightarrow \text{for initial state} \quad (1)$$

$$q_2 = q_1 1 + q_2 1 + q_3 1 \longrightarrow (2)$$

$$q_3 = q_2 0 \longrightarrow (3)$$

From (2),

$$q_2 = q_1 1 + q_2 1 + q_3 1$$

$$q_2 = q_1 1 + q_2 1 + q_2 01$$

$$\underbrace{q_2}_{R} = \underbrace{q_1 1}_{Q} + \underbrace{q_2}_{R} \underbrace{(1+01)}_{P}$$

Now $R = Q + RP$ can be replaced with $R = QP^*$

$$q_2 = q_1 1 (1+01)^* \longrightarrow (4)$$

Now $q_3 = q_2 0$

Put value of q_2 from (4)

$$q_3 = q_1 1 (1+01)^* 0 \longrightarrow (5)$$

Now from (1),

$$q_1 = q_1 0 + q_3 0 + \epsilon$$

$$q_1 = q_1 0 + q_1 1 (1+01)^* 00 + \epsilon \quad [\text{From } (5)]$$

$$\underbrace{q_1}_{R} = \underbrace{\epsilon}_{Q} + \underbrace{q_1}_{R} \underbrace{(0 + 1(1+01)^* 00)}_{P}$$

$$q_1 = \epsilon (0 + 1(1+01)^* 00)^*$$

$$q_1 = (0 + 1(1+01)^* 00)^*$$

is required RE

Q Convert given FA to RE

Ans

$$q_1 = q_1 0 + \epsilon \rightarrow (1)$$

$$q_2 = q_1 1 + q_2 1 \rightarrow (2)$$

$$q_3 = q_2 0 + q_3 0 + q_3 1 \rightarrow (3)$$

Now (3),

$$q_3 = q_2 0 + q_3 0 + q_3 1$$

$$\underset{R}{q_3} = \underset{Q}{q_2 0} + \underset{R}{q_3}\underbrace{(0+1)}_{P}$$

$$q_3 = q_2 0 (0+1)^*$$

→ No need to simplify q_3 as we need solution ~~to~~ of final states

Now from (1),

$$\underset{R}{q_1} = \underset{Q}{\epsilon} + \underset{R}{q_1} \underset{P}{0}$$

$$q_1 = \epsilon 0^*$$

$$\boxed{q_1 = 0^*}$$

Now eq. (2),

$$\underset{R}{q_2} = \underset{Q}{q_1 1} + \underset{R}{q_2} \underset{P}{1}$$

$$q_2 = q_1 1 (1)^*$$

Put value of q_1

$$\boxed{q_2 = 0^*11^*}$$

~~0^*11^* is required RE~~

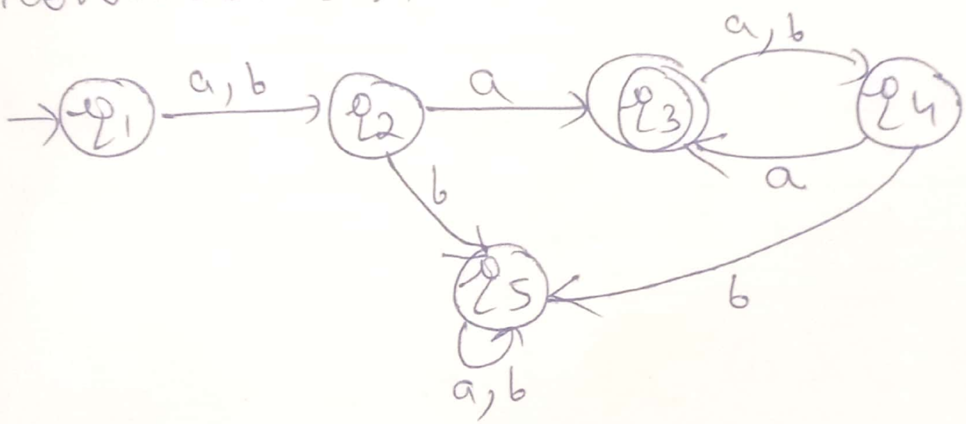
$q_1 + q_2$ is required RE

$= 0^* + 0^*11^*$

$= 0^*(\epsilon + 11^*)$ $\qquad \because PP^* = P^*$

$= 0^*1^*$

Q Convert DFA to RE



Ans

$q_1 = \epsilon \rightarrow (1)$

$q_2 = q_1 a + q_1 b \rightarrow (2)$

$q_3 = q_2 a + q_4 a \rightarrow (3)$

$q_4 = q_3 a + q_3 b \rightarrow (4)$

$q_5 = q_2 b + q_5 a + q_5 b + q_4 b \rightarrow (5)$

From (2)

$q_2 = q_1 a + q_1 b$

$\quad = \epsilon a + \epsilon b \quad = a + b$

$q_3 = q_2 a + q_4 a$

$q_3 = (a+b)a + q_4 a \rightarrow (6)$

Now,

$q_4 = q_3(a+b)$

Put in (6)

$\underbrace{q_3}_{R} = \underbrace{(a+b)a}_{Q} + \underbrace{q_3}_{R}\underbrace{(a+b)a}_{P}$

$\boxed{q_3 = (a+b)a((a+b)a)^*}$

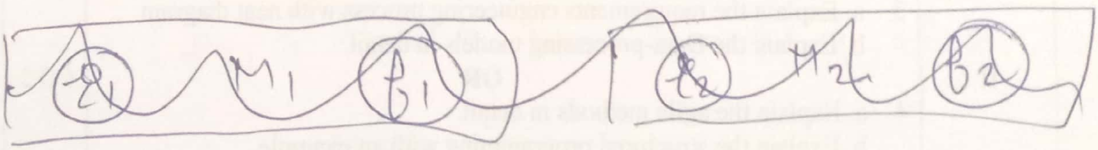
Since q_3 is final state, it is required RE.

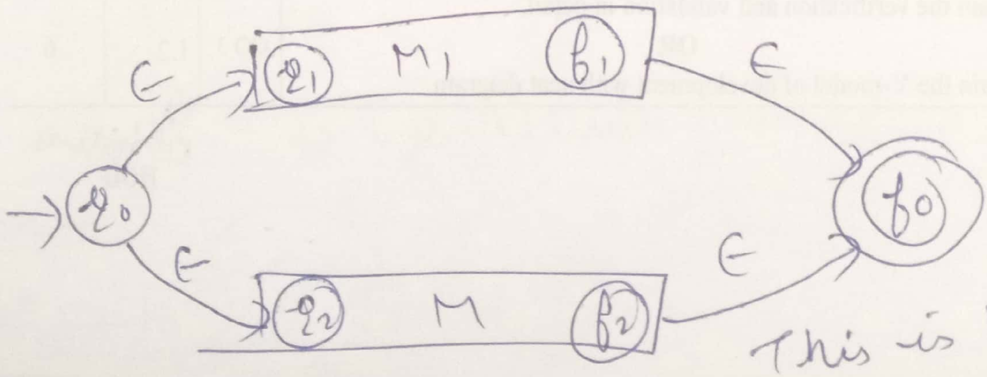
Convert from RE to FA

Union → Let r_1 & r_2 be RE having FA i.e FA_1 & FA_2, then their union also have FA, M which accepts both r_1 & r_2

$FA_1 = (Q_1, \Sigma_1, S_1, q_1, f_1)$

$FA_2 = (Q_2, \Sigma_2, S_2, q_2, f_2)$





This is FA for M.

$M = ((Q_1 \cup Q_2 \cup q_0 \cup f_0), \Sigma_1 \cup \Sigma_2, \delta, q_0, f_0)$

Now,

$\delta(q_0, \epsilon) = \{q_1, q_2\}$

$\delta(q, a) = \delta_1(q, a)$ with q in $Q_1 - \{f_1\}$, $a \in \Sigma_1 \cup \{\epsilon\}$

$\delta(q, a) = \delta_2(q, a)$ when q in $Q_2 - \{f_2\}$, $a \in \Sigma_2 \cup \{\epsilon\}$

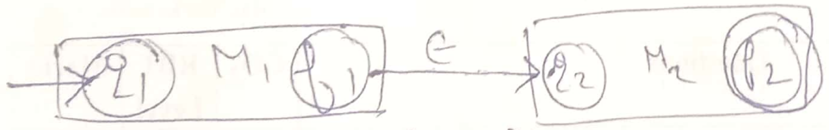
$\delta(f_1, \epsilon) = \cancel{\phi}\ \delta(f_2, \epsilon) = f_0$

<u>concatenation</u>

For r_1, M_1 is FA

For r_2, M_2 is FA

For $r_1 r_2$, M is is FA



This is FA for M

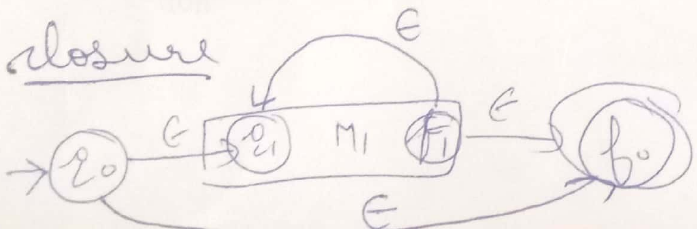
$M = (Q_1 \cup Q_2, \Sigma_1 \cup \Sigma_2 \cup \epsilon, \delta, q_1, f_2)$

$\delta(q, a) = \delta_1(q_1, a)$ with q in $Q_1 - \{f_1\}$, $a \in \Sigma_1 \cup \{\epsilon$

$\delta(f_1, \epsilon) = q_2$

$\delta(q, a) = \delta_2(q_1, a)$ with q in Q_2, $a \in \Sigma_2 \cup \{\epsilon\}$

<u>closure</u>



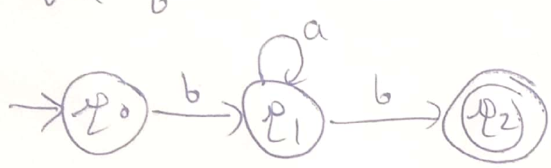
$\delta(q_0, \epsilon) = \{q_1, f_0\}$

$\delta(q, a) = \delta_1(q, a) \; \forall q$ in $Q_1 - \{f_1\}$ & $a \in \Sigma_1 \cup \{\epsilon\}$

$\delta(f_1, \epsilon) = \{f_0, q_1\}$

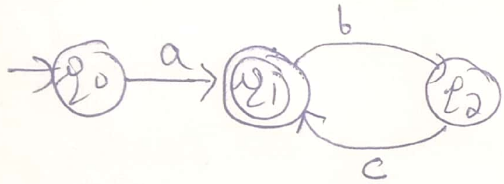
Q Convert RE to FA

Ans a) ba^*b

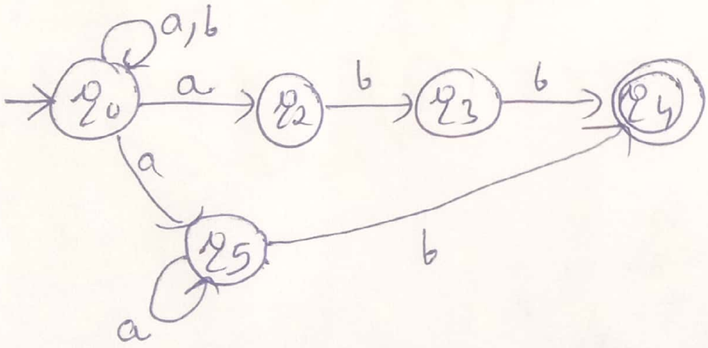


b) $(a+b)c$

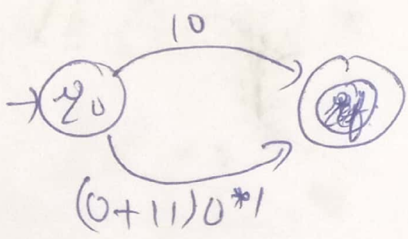
c) $a(bc)^*$

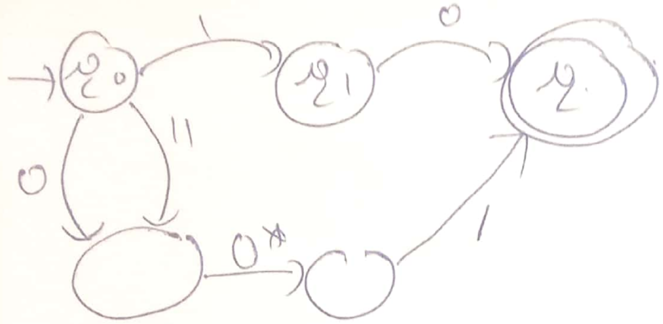


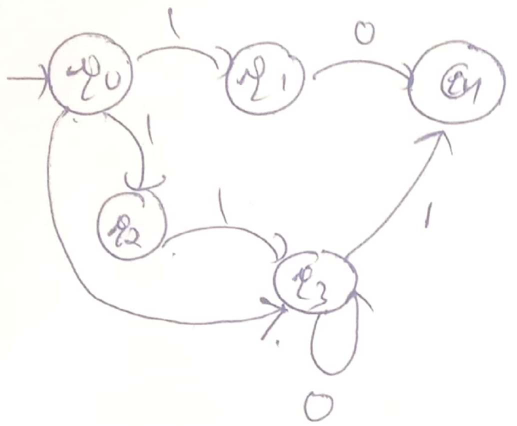
d) $\underbrace{(a|b)^*}_{R_1}\underbrace{(abb|a^+b)}_{R_2}$



e) $10+(0+11)0^*1$







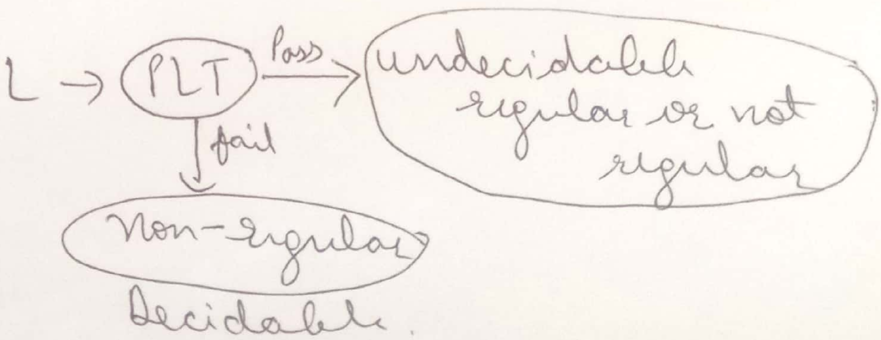
Pumping Lemma

If L is an infinite language then there exists some ~~position~~ positive integer 'n' (pumping length) such that any string $w \in L$ has length greater than or equal to 'n' i.e. $|w| \geq n$ then w can be divided into 3 parts, $w = xyz$ satisfying the following conditions

(i) for each $i \geq 0$, $xy^iz \in L$

(ii) $|y| > 0$ and

(iii) $|xy| \leq n$



eg - $L = \{a^n b^n ; n \geq 0\}$

$\underbrace{aa}_{x}\underbrace{bb}_{y}\underbrace{bb}_{z} \in L$

$\underline{aa}_{x}\ \underline{bbbb}_{y}\ \underline{bb}_{z} \notin L$

Thus, language is not regular.

$\underbrace{a}_{x}\underbrace{ab}_{y}\underbrace{bbb}_{z} \in L$

$\underbrace{a}_{x}\ \underbrace{abab}_{y}\ \underbrace{bbb}_{z} \notin L$

Not regular

Applications

① It is to be applied to show that certain languages are not regular. It should never be used to show a language is regular

→ if L is regular, ~~it satisfies~~ pumping lemma

→ if L does not satisfy pumping lemma, it is non-regular

Q Prove that $L = \{a^i b^i \mid i \geq 0\}$ is not regular

Closure properties of Regular Language

1. Union $(L_1 \cup L_2)$
2. Concatenation $(L_1 L_2)$
3. Closure $(*)$ L^*
4. Complementation $\bar{L} = \Sigma^* - L$
5. Intersection $L1 \cap L2 = \overline{\bar{L1} \cup \bar{L2}}$
6. Difference $L1 - L2 = L_1 \cap \bar{L_2}$
7. Reversal $(L)^R$
8. Homomorphism
9. Reverse homomorphism
10. Quotient
11. INIT
12. Substitution

closed means if L1 & L2 are regular then L1 ⊕ L2 is also regular. It means they are closed under ⊕

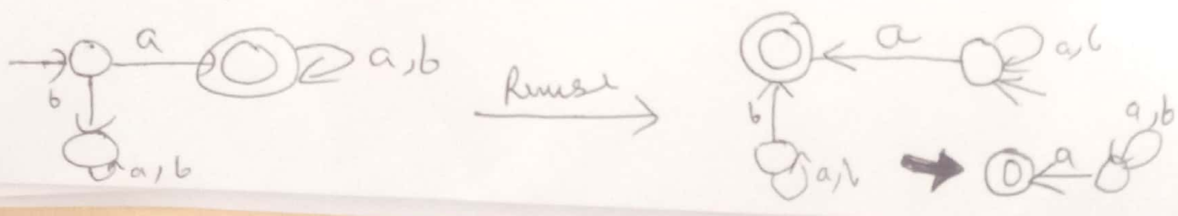
13. Infinite union → not closed

Regular language are closed under reversal

1. Make initial state of M as final state of M'
2. Final state of M become initial state of M'
3. Reverse the direction of edges of M to make M'
4. No change in loop & remove unnecessary states

eg- L_1 = {set of all strings over (a,b) starting with a}

$a(a+b)^*$



Init operation (initial / prefix)

set of all prefix of $w \in L$

Let $L = \{ab, ba\}$

prefix of ab $= \epsilon, a, ab$

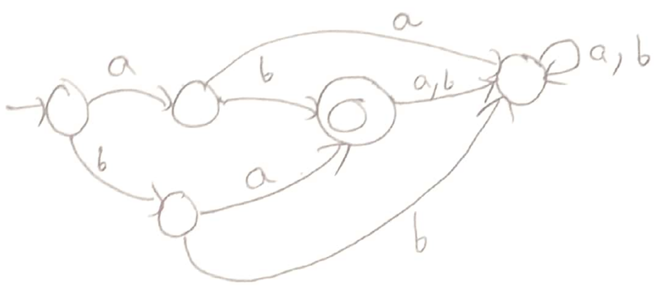
" " abc $= \epsilon, a, ab, abc$

" " ba $= \epsilon, b, ba$

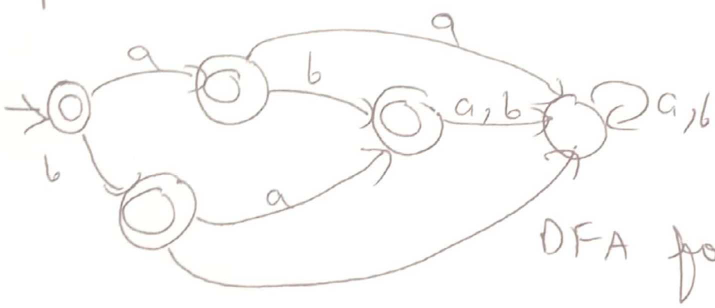
init of $L = \{\epsilon, a, b, ab, ba\}$

Regular languages are closed under init.

$L = \{ab, ba\}$



Make all non-final states as final state except trap state



DFA for $L = \{\epsilon, a, b, ab, ba\}$
init operation of $L = \{ab, ba\}$

If we can make DFA for init operation then we can say that RL are closed under init operation

Regular languages are not closed under infinite union

L1UL2 → regular

$L_1 = \{a^1 b^1\} \cup$

$L_2 = \{a^2 b^2\} \cup$

$L_3 = \{a^3 b^3\} \cup$

$L_4 = \{a^4 b^4\} \cup$

⋮

$a^n b^n$ where $n \geq 1$ (infinite union)

Now $L = \{a^n b^n$ where $n \geq 1\}$ is not regular

So, RL are not closed under infinite union

Closure of Languages

	Regular	DCFL	CFL	CSL	REC	RE
$\cup$	Yes	No	Yes	Yes	Yes	Yes
$\cap$	Yes	No	No	Yes	Yes	Yes
L^c	Yes	Yes	No	Yes	Yes	No
$\cdot$	Yes	No	Yes	Yes	Yes	Yes
$*$	Yes	No	Yes	Yes	Yes	Yes
$+$	Yes	No	Yes	Yes	Yes	Yes

REC → recursive language

RE → recursively enumerable

DCFL → Deterministic CFL

CFL → context free language

CSL → context sensitive language.